

Lab Report: Istio Getting Started on Minikube

Student Name: Lu Weichi

Course: Cloud Computing

Date: 2026-05-16

Tools Used: Minikube, kubectl, istioctl, Docker Desktop, WSL (Ubuntu), Git Bash

Overview

This lab follows the official Istio Getting Started guide. The main objectives are:

Install Istio control plane on a Minikube cluster.

Install Kubernetes Gateway API CRDs.

Deploy the Bookinfo sample application with automatic sidecar injection.

Expose the application outside the cluster using an Istio Gateway.

Access the application via port-forwarding and verify its functionality.

During the lab, several network-related issues occurred (image pull failures). They were resolved by manually pulling images from a domestic mirror (docker.m.daocloud.io) and loading them into the Minikube environment.

Procedure

Step 1 – Download and Install Istio

1.1 Download the Istio release

```
curl -L https://istio.io/downloadIstio | sh -
```

1.2 Move to the Istio directory and add istioctl to PATH

```
cd istio-1.29.2 & export PATH=$PWD/bin:$PATH
```

1.3 Manually pull the istio/pilot:1.29.2 image from a domestic mirror

```
docker pull docker.m.daocloud.io/istio/pilot:1.29.2
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/pilot:1.29.2
1.29.2: Pulling from istio/pilot
8c7170db3ec9: Pull complete
689b91d88a0f: Pull complete
ede6d5e95b67: Pull complete
dec211a11132: Pull complete
Digest: sha256:3681fe197870abad5c953107e241e104a813ba2d6e639333ff53e0545348ff8b1
Status: Downloaded newer image for docker.m.daocloud.io/istio/pilot:1.29.2
docker.m.daocloud.io/istio/pilot:1.29.2
```

1.4 Verify that the image has been pulled and re-tag it

`docker` images | `grep` istio/pilot

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker images | grep istio/pilot
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
docker.m.daocloud.io/istio/pilot:1.29.2      3681fe197870      310MB      73.6MB
istio/pilot:1.29.2                          3681fe197870      310MB      73.6MB
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/pilot:1.29.2
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl delete namespace istio-system --ignore-not-found=true
namespace "istio-system" deleted
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ sleep 5
```

1.5 Load the image into the Minikube internal Docker daemon

`minikube image load istio/pilot:1.29.2`

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/pilot:1.29.2
```

1.6 Remove any previous failed Istio namespace (to ensure a clean installation)

`kubectl delete namespace istio-system --ignore-not-found=true`

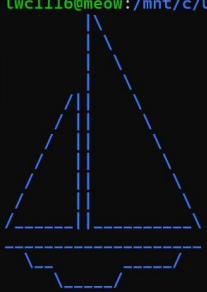
`sleep 5`

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl delete namespace istio-system --ignore-not-found=true
namespace "istio-system" deleted
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ sleep 5
```

1.7 Install Istio using the demo profile (without default gateways)

`istioctl install -f samples/bookinfo/demo-profile-no-gateways.yaml -y`

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ istioctl install -f samples/bookinfo/demo-profile-no-gateways.yaml -y
```



```
✓ Istio core installed 🚢
✓ Istiod installed 🌸
✓ Installation complete
```

Step 2 – Install Kubernetes Gateway API CRDs

The Gateway API CRDs are not installed by default in Minikube. Install them using the official YAML manifest:

`kubectl apply -f https://github.com/kubernetes-sigs/gateway-api/releases/download/v1.1.0/standard-install.yaml`

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/gateway-api/bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
```

Verify the CRDs are registered:

`kubectl get crd | grep gateway.networking.k8s.io`

(No screenshot provided for this verification command, but it can be assumed successful.)

Step 3 – Deploy the Bookinfo Sample Application

3.1 Initial deployment

```
bash
```

```
kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods
No resources found in default namespace.
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
details-v1-6cc9f5cc44-656br         0/2     Init:ImagePullBackOff 0            87s
productpage-v1-7f885b46fc-n6xss     0/2     Init:ErrImagePull     0            87s
ratings-v1-77b8b6df5b-24725        0/2     Init:ImagePullBackOff 0            87s
reviews-v1-fdbf79cd8-6x2fj          0/2     Init:ImagePullBackOff 0            87s
reviews-v2-674c6d8b4-t8rmg          0/2     Init:ImagePullBackOff 0            87s
reviews-v3-7b775c7568-jjslx        0/2     Init:ImagePullBackOff 0            87s
```

3.2 Pods stuck in ImagePullBackOff

After deployment, all pods were stuck in `Init:ImagePullBackOff` or `ErrImagePull` because the required images could not be pulled from Docker Hub.

```
kubectl get pods
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
details-v1-6cc9f5cc44-656br         0/2     Init:ImagePullBackOff 0            87s
productpage-v1-7f885b46fc-n6xss     0/2     Init:ErrImagePull     0            87s
ratings-v1-77b8b6df5b-24725        0/2     Init:ImagePullBackOff 0            87s
reviews-v1-fdbf79cd8-6x2fj          0/2     Init:ImagePullBackOff 0            87s
reviews-v2-674c6d8b4-t8rmg          0/2     Init:ImagePullBackOff 0            87s
reviews-v3-7b775c7568-jjslx        0/2     Init:ImagePullBackOff 0            87s
```

Check which images are needed:

```
kubectl describe pod details-v1-6cc9f5cc44-656br | grep -A 1 "Image:"
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl describe pod details-v1-6cc9f5cc44-656br | grep -A 1 "Image:"
  Image:          docker.io/istio/proxyv2:1.29.2
  Image ID:
--
  Image:          docker.io/istio/proxyv2:1.29.2
  Image ID:
--
  Image:          docker.io/istio/examples-bookinfo-details-v1:1.20.3
  Image ID:
```

3.3 Manually pull images from a domestic mirror and load into Minikube

Pull the sidecar image `proxyv2`

```
docker pull docker.m.daocloud.io/istio/proxyv2:1.29.2docker tag docker.m.daocloud.io/istio/proxyv2:1.29.2 istio/proxyv2:1.29.2minikube image load istio/proxyv2:1.29.2
```

it was done earlier in Step 1.

Pull all Bookinfo application images (tag 1.20.3)

For each image, the same pattern was used: `docker pull`, `docker tag`, `minikube image load`.

Example – details-v1

bash

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
```

```
docker tag docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3 istio/examples-bookinfo-details-v1:1.20.3minikube image load istio/examples-bookinfo-details-v1:1.20.3
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-details-v1
c913caa0e3af: Download complete
8a628cdd7ccc: Pull complete
bb9880ce56e8: Pull complete
499e3cd3c5ce: Pull complete
87b7f3d9b8a5: Pull complete
e679989898c6: Pull complete
688113cd94a2: Pull complete
9599f7ae6bd0: Pull complete
7d982d31d2b7: Pull complete
6f7d3ac89af3: Pull complete
Digest: sha256:a49731e1fc05d7c24a4103709f41d64180bed989b251625f4c31ba63022e303b
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
istio/examples-bookinfo-details-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-details-v1:1.20.3
```

Example – ratings-v1

bash

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3docker tag ... istio/examples-bookinfo-ratings-v1:1.20.3minikube image load
```

...

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-ratings-v1
58e674205d2a: Pull complete
4f4fb700ef54: Pull complete
954666baad70: Pull complete
241b31739b9c: Pull complete
1667adbcc2a4: Pull complete
af8df1bc3450: Pull complete
96ada0c95375: Pull complete
ee293fb0644f: Pull complete
809b04e40f4e: Pull complete
7906dc411a8e: Download complete
Digest: sha256:aad8f7bb46664704eecd92d9ebe8563e3e41571e2301f593cc521301b14ae51d
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
istio/examples-bookinfo-ratings-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-ratings-v1:1.20.3
```

Example – reviews-v1

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v1:1.20.3docker tag ... istio/examples-bookinfo-reviews-v1:1.20.3minikube image load
d ...
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v1:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-reviews-v1
738ae7fc37c1: Pull complete
f5573ccbc8d8: Pull complete
458449456871: Pull complete
ec858451a832: Pull complete
ee075a90a47e: Pull complete
30a9c22ae099: Pull complete
aae5db5a1a6f: Pull complete
099deadc3548: Pull complete
14c1757fcd37: Pull complete
53b0ec548778: Pull complete
87b68f04c9df: Pull complete
c8cb19475be1: Pull complete
897404661be1: Pull complete
c240b66c3900: Pull complete
71b195ed4caf: Download complete
Digest: sha256:e2efe32bf087dcd35c397fc6766c0bdcbae77855df7919377bcf047e1a6ded12
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-reviews-v1:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-reviews-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-reviews-v1:1.20.3
istio/examples-bookinfo-reviews-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-reviews-v1:1.20.3
```

Example – reviews-v2

bash

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v2:1.20.3docker tag ... istio/examples-bookinfo-reviews-v2:1.20.3minikube image loa
d ...
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v2:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-reviews-v2
2a07348fa321: Download complete
Digest: sha256:09b92ca16738b5bde2ae66ba0a4b21d32e86d8b6ddb34331abfe1179d8c6298
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-reviews-v2:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-reviews-v2:1.20.3
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-reviews-v2:1.20.3
istio/examples-bookinfo-reviews-v2:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-reviews-v2:1.20.3
```

Example – reviews-v3

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v3:1.20.3docker tag ... istio/examples-bookinfo-reviews-v3:1.20.3minikube image loa
d ...
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-reviews-v3:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-reviews-v3
fa8f51bd21cf: Download complete
Digest: sha256:3b3415c5e7cc7cef3ad1741fa595f2f26627bddf2117dcee1429fa2a422df1b7
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-reviews-v3:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-reviews-v3:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-reviews-v3:1.20.3
istio/examples-bookinfo-reviews-v3:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-reviews-v3:1.20.3
```

Example – productpage-v1

```
docker pull docker.m.daocloud.io/istio/examples-bookinfo-productpage-v1:1.20.3docker tag ... istio/examples-bookinfo-productpage-v1:1.20.3minikube i
mage load ...
```



```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-productpage-v1:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-productpage-v1
f89b8a1b3342: Pull complete
478b8250c2cd: Pull complete
9c5eda75c785: Pull complete
8171fb1ef23b: Pull complete
adf7749ab929: Pull complete
4f4fb700ef54: Pull complete
120e9bbdebeb: Pull complete
abbbb0896802: Pull complete
72d67fbb9156: Pull complete
29984f11a3e9: Pull complete
0f1e2b67e655: Pull complete
d61f04621922: Pull complete
be94d3be81c7: Pull complete
db42ddceabbc: Pull complete
1bc22db744b6: Download complete
Digest: sha256:1e339866b782d22b063539d19c87b4682291c889e67c3d4abdea947bef917958
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-productpage-v1:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-productpage-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-productpage-v1:1.20.3 istio/examples-bookinfo-productpage-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-productpage-v1:1.20.3
```

3.4 Redeploy the application after loading all images

Delete the old deployments:

```
kubectl delete deployment --all -n default
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl delete deployment --all -n default
deployment.apps "details-v1" deleted from default namespace
deployment.apps "productpage-v1" deleted from default namespace
deployment.apps "ratings-v1" deleted from default namespace
deployment.apps "reviews-v1" deleted from default namespace
deployment.apps "reviews-v2" deleted from default namespace
deployment.apps "reviews-v3" deleted from default namespace
```

Apply the Bookinfo configuration again:

```
kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
service/details unchanged
serviceaccount/bookinfo-details unchanged
deployment.apps/details-v1 created
service/ratings unchanged
serviceaccount/bookinfo-ratings unchanged
deployment.apps/ratings-v1 created
service/reviews unchanged
serviceaccount/bookinfo-reviews unchanged
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage unchanged
serviceaccount/bookinfo-productpage unchanged
deployment.apps/productpage-v1 created
```

3.5 Verify all pods are running

```
bash
```

```
kubectl get pods -w
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
details-v1-6cc9f5cc44-gj54v        2/2     Running   0           9s
productpage-v1-7f885b46fc-vllbf    2/2     Running   0           9s
ratings-v1-77b8b6df5b-fdd6j        2/2     Running   0           9s
reviews-v1-fdbf79cd8-mwrvb         2/2     Running   0           9s
reviews-v2-674c6d8b4-hfrd5         2/2     Running   0           9s
reviews-v3-7b775c7568-pfvwt        2/2     Running   0           9s
```

Verify that each pod has the sidecar injected (two containers):

```
kubectl describe pod details-v1-6cc9f5cc44-gj54v | grep -A 1 "Image:"
```

Step 4 – Open the Application to Outside Traffic (Create Gateway)

4.1 Create a Kubernetes Gateway and HTTPRoute

bash

```
kubectl apply -f samples/bookinfo/gateway-api/bookinfo-gateway.yaml
```

```
^Clwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/gateway-api/bookinfo-gateway.yaml
resource mapping not found for name: "bookinfo-gateway" namespace: "" from "samples/bookinfo/gateway-api/bookinfo-gateway.yaml": no matches for kind "Gateway" in version "gateway.networking.k8s.io/v1"
ensure CRDs are installed first
resource mapping not found for name: "bookinfo" namespace: "" from "samples/bookinfo/gateway-api/bookinfo-gateway.yaml": no matches for kind "HTTPRoute" in version "gateway.networking.k8s.io/v1"
ensure CRDs are installed first
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get crd gateways.gateway.networking.k8s.io &> /dev/null || \
{ kubectl kustomize "github.com/kubernetes-sigs/gateway-api/config/crd?ref=v1.1.0" | kubectl apply -f -; }
error: hit 27s timeout running '/usr/bin/git fetch --depth=1 https://github.com/kubernetes-sigs/gateway-api v1.1.0'
error: no objects passed to apply
```

4.2 Change the gateway service type to ClusterIP (required for Minikube)

```
kubectl annotate gateway bookinfo-gateway networking.istio.io/service-type=ClusterIP --namespace default
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl annotate gateway bookinfo-gateway networking.istio.io/service-type=ClusterIP --namespace default
gateway.gateway.networking.k8s.io/bookinfo-gateway annotated
```

Step 5 – Access the Application

5.1 Port-forward the gateway service to localhost

```
kubectl port-forward svc/bookinfo-gateway-istio 8080:80 &
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl port-forward svc/bookinfo-gateway-istio 8080:80 &
[1] 14521
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ Forwarding from 127.0.0.1:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

5.2 Open the Bookinfo web page

In a browser, navigate to <http://localhost:8080/productpage>.

【截图：浏览器显示 Bookinfo 应用首页，包含 “The Comedy of Errors” 书籍信息、书评和星级，底部显示 “Reviews

localhost:8080/productpage

aa

☆

The Comedy of Errors

Wikipedia Summary: The Comedy of Errors is one of **William Shakespeare's** early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

[Learn more about Istio →](#)

Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"



Reviewer1

Reviews served by: reviews-v3-7b775c7568-pfvvt

★★★★☆

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."



Reviewer2

Reviews served by: reviews-v3-7b775c7568-pfvvt

Troubleshooting Summary

During the lab, several issues were encountered due to network restrictions (inability to directly access Docker Hub and Google Container Registry). Each problem was resolved manually as described below.

1. Istio installation failed because the `istio/pilot:1.29.2` image could not be pulled.

Symptom:

`istioctl install` timed out after 5 minutes with an error: `ImagePullBackOff: Back-off pulling image "docker.io/istio/pilot:1.29.2".`


```

lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods
No resources found in default namespace.
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
details-v1-6cc9f5cc44-656br	0/2	Init:ImagePullBackOff	0	87s
productpage-v1-7f885b46fc-n6xss	0/2	Init:ErrImagePull	0	87s
ratings-v1-77b8b6df5b-24725	0/2	Init:ImagePullBackOff	0	87s
reviews-v1-fdbf79cd8-6x2fj	0/2	Init:ImagePullBackOff	0	87s
reviews-v2-674c6d8b4-t8rmg	0/2	Init:ImagePullBackOff	0	87s
reviews-v3-7b775c7568-jjslx	0/2	Init:ImagePullBackOff	0	87s

Cause:

Direct access to docker.io was blocked or extremely slow in the local network environment.

Solution:

The image was manually pulled from a domestic mirror (docker.m.daocloud.io), re-tagged, and loaded into the Minikube internal Docker daemon.

```
docker pull docker.m.daocloud.io/istio/pilot:1.29.2
docker tag docker.m.daocloud.io/istio/pilot:1.29.2 istio/pilot:1.29.2
minikube image load istio/pilot:1.29.2
```

After this, `istioctl install` completed successfully.

2. Bookinfo pods stuck in ImagePullBackOff after deployment.

Symptom:

Running `kubectl get pods` showed all pods with `READY 0/2` and status `Init:ImagePullBackOff` or `ErrImagePull`.

```

lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/proxyv2:1.29.2
1.29.2: Pulling from istio/proxyv2
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://image-mirror.r2.daocloud.vip/docker/registry/v2/blobs/sha256/c9/c99272fbb1fc7a39091da7aeff63d16dfdb42e204acf8e23a69dc1a75790d2/data?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=420d2a5e9cfd0dcddae7cfa0f609cd87%2F20260515%2Fapac%2Fs3%2Faws4_request&X-Amz-Date=20260515T161247Z&X-Amz-Expires=86400&X-Amz-SignedHeaders=host&X-Amz-Signature=b096aaf0cca25633d327f555479217b79cc8e8d3649c7a5658fd535a3b239058": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/proxyv2:1.29.2
1.29.2: Pulling from istio/proxyv2
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://image-mirror.r2.daocloud.vip/docker/registry/v2/blobs/sha256/c9/c99272fbb1fc7a39091da7aeff63d16dfdb42e204acf8e23a69dc1a75790d2/data?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=420d2a5e9cfd0dcddae7cfa0f609cd87%2F20260515%2Fapac%2Fs3%2Faws4_request&X-Amz-Date=20260515T161323Z&X-Amz-Expires=86400&X-Amz-SignedHeaders=host&X-Amz-Signature=cc553dc1d1e144f36a612c4f5962e0e9349fdf274d80d63a14bac41cba8b9fc": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull registry.cn-hangzhou.aliyuncs.com/google_containers/proxyv2:1.29.2
Error response from daemon: pull access denied for registry.cn-hangzhou.aliyuncs.com/google_containers/proxyv2, repository does not exist or may require 'docker login'
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.rainbond.cc/istio/proxyv2:1.29.2
Error response from daemon: failed to resolve reference "docker.rainbond.cc/istio/proxyv2:1.29.2": failed to do request: Head "https://docker.rainbond.cc/v2/istio/proxyv2/manifests/1.29.2": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull hub.rat.dev/istio/proxyv2:1.29.2
Error response from daemon: failed to resolve reference "hub.rat.dev/istio/proxyv2:1.29.2": failed to do request: Head "https://hub.rat.dev/v2/istio/proxyv2/manifests/1.29.2": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.rainbond.cc/istio/proxyv2:1.29.2
Error response from daemon: failed to resolve reference "docker.rainbond.cc/istio/proxyv2:1.29.2": failed to do request: Head "https://docker.rainbond.cc/v2/istio/proxyv2/manifests/1.29.2": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.rainbond.cc/istio/proxyv2:1.29.2
Error response from daemon: failed to resolve reference "docker.rainbond.cc/istio/proxyv2:1.29.2": failed to do request: Head "https://docker.rainbond.cc/v2/istio/proxyv2/manifests/1.29.2": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.rainbond.cc/istio/proxyv2:1.29.2
Error response from daemon: failed to resolve reference "docker.rainbond.cc/istio/proxyv2:1.29.2": failed to do request: Head "https://docker.rainbond.cc/v2/istio/proxyv2/manifests/1.29.2": EOF
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.19.0
1.19.0: Pulling from istio/examples-bookinfo-details-v1
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://image-mirror.r2.daocloud.vip/docker/registry/v2/blobs/sha256/59/599a33dae563796cd16beca8320791db2aa09a8c16b269ced3fc59fd3f151fca/data?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=420d2a5e9cfd0dcddae7cfa0f609cd87%2F20260515%2Fapac%2Fs3%2Faws4_request&X-Amz-Date=20260515T162304Z&X-Amz-Expires=86400&X-Amz-SignedHeaders=host&X-Amz-Signature=adf881abd577974ebe303e3f0f0fdd50aa6db0c8f97ee6111f1a57a804b94fd": net/http: TLS handshake timeout

```

Cause:

Both the Istio sidecar image (istio/proxyv2:1.29.2) and the six Bookinfo application images (e.g., examples-bookinfo-details-v1:1.20.3) could not be pulled from Docker Hub.

Solution:

I changed the network node of the VPN. Each missing image was manually pulled from [docker.m.daocloud.io](https://m.daocloud.io/), re-tagged, and loaded into Minikube. The required application images were:

[examples-bookinfo-details-v1:1.20.3](https://m.daocloud.io/examples-bookinfo-details-v1:1.20.3)

[examples-bookinfo-ratings-v1:1.20.3](https://m.daocloud.io/examples-bookinfo-ratings-v1:1.20.3)

[examples-bookinfo-reviews-v1:1.20.3](https://m.daocloud.io/examples-bookinfo-reviews-v1:1.20.3)

[examples-bookinfo-reviews-v2:1.20.3](https://m.daocloud.io/examples-bookinfo-reviews-v2:1.20.3)

[examples-bookinfo-reviews-v3:1.20.3](https://m.daocloud.io/examples-bookinfo-reviews-v3:1.20.3)

[examples-bookinfo-productpage-v1:1.20.3](https://m.daocloud.io/examples-bookinfo-productpage-v1:1.20.3)

Example commands for proxyv2 and one application image:

```
# Sidecar docker pull docker.m.daocloud.io/istio/proxyv2:1.29.2 docker tag docker.m.daocloud.io/istio/proxyv2:1.29.2 istio/proxyv2:1.29.2minikube image load istio/proxyv2:1.29.2
```

```
# Application image (details) docker pull docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3
```

```
docker tag docker.m.daocloud.io/istio/examples-bookinfo-details-v1:1.20.3 istio/examples-bookinfo-details-v1:1.20.3minikube image load istio/examples-bookinfo-details-v1:1.20.3
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker pull docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
1.20.3: Pulling from istio/examples-bookinfo-ratings-v1
58e674205d2a: Pull complete
4f4fb700ef54: Pull complete
954666baad70: Pull complete
241b31739b9c: Pull complete
1667adbcc2a4: Pull complete
af8df1bc3450: Pull complete
96ada0c95375: Pull complete
ee293fb0644f: Pull complete
809b04e40f4e: Pull complete
7906dc411a8e: Download complete
Digest: sha256:aad8f7bb46664704eecd92d9ebe8563e3e41571e2301f593cc521301b14ae51d
Status: Downloaded newer image for docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ docker tag docker.m.daocloud.io/istio/examples-bookinfo-ratings-v1:1.20.3
istio/examples-bookinfo-ratings-v1:1.20.3
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ minikube image load istio/examples-bookinfo-ratings-v1:1.20.3
```

After loading all images, the deployment was re-applied. All pods then became **Running** with **READY 2/2**.

3. Gateway creation failed with “no matches for kind Gateway”.

Symptom:

Executing `kubectl apply -f samples/bookinfo/gateway-api/bookinfo-gateway.yaml` produced:

```
error: no matches for kind "Gateway" in version "gateway.networking.k8s.io/v1"
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl get crd gateways.gateway.networking.k8s.io && /dev/null || \
{ kubectl kustomize "github.com/kubernetes-sigs/gateway-api/config/crd?ref=v1.1.0" | kubectl apply -f -; }
error: hit 27s timeout running '/usr/bin/git fetch --depth=1 https://github.com/kubernetes-sigs/gateway-api v1.1.0'
error: no objects passed to apply
```

Cause:

The Kubernetes Gateway API Custom Resource Definitions (CRDs) were not installed in the cluster.

Solution:

The official Gateway API CRDs were installed using the standard YAML manifest:

```
kubectl apply -f https://github.com/kubernetes-sigs/gateway-api/releases/download/v1.1.0/standard-install.yaml
```

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f https://github.com/kubernetes-sigs/gateway-api/releases/download/v1.1.0/standard-install.yaml
customresourcedefinition.apiextensions.k8s.io/gatewayclasses.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gateways.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/grpcroutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/httproutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/referencegrants.gateway.networking.k8s.io created
```

After the CRDs were installed, the `bookinfo-gateway.yaml` applied successfully.

```
lwc1116@meow:/mnt/c/Users/21315/istio-1.29.2$ kubectl apply -f samples/bookinfo/gateway-api/bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
```

All issues were successfully resolved.

Conclusion

In this lab, I successfully installed Istio on a Minikube cluster, deployed the Bookinfo sample application with automatic sidecar injection, exposed it using an Istio Gateway, and accessed it via port-forwarding. Although network difficulties required manual intervention (pulling images from a domestic mirror), the final result demonstrates a fully functional service mesh. The lab confirmed Istio’s ability to manage traffic, inject sidecars, and provide external access through Gateway API.